

Tyco Brahe Mera

CURRICULUM VITAE

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EDUCATION

- Florida State University — Ph.D., Astrophysics** Tallahassee, FL | Aug 2022 – May 2026
Thesis: *Probing the 3D Structures of Supernovae through Molecules*
Advisor: Peter Hoefflich
- Florida State University — M.S., Physics** Tallahassee, FL | Aug 2020 – Aug 2022
- Brown University — Sc.B., Honors, Astrophysics** Providence, RI | Aug 2016 – May 2020
Thesis: *A Pedagogical Demonstration/Lecture Plan for the Introduction of Gravitational Wave Astronomy*
Advisor: Robert Coyne

EMPLOYMENT

- University of Hawai'i at Manoa — Postdoctoral Researcher** Honolulu, HI | Jul 2026 – Present
- Studying three-dimensional structures of core-collapse supernovae through molecular spectroscopy.
 - Developing shock interaction models for SNe Ia-CSM, Type II SNe, and Little Red Dots.
 - Developing dust models using the same framework as the molecular modeling developed at Florida State University.
 - Extending molecular radiation transport models to include directional dependence and viewing-angle effects.
- Florida State University — Graduate Research Assistant** Tallahassee, FL | Aug 2020 – May 2026
- Conducted theoretical and computational astrophysics research focused on probing the three-dimensional structures of core-collapse supernovae through molecular spectroscopy.
 - Developed computational modeling tools and analyzed JWST observations.
 - Performed high-performance computing simulations and published peer-reviewed research.
- Los Alamos National Laboratory — Graduate Research Assistant** Los Alamos, NM | Jul 2022 – Aug 2023
- Integrated a second-order fast sweeping method for the Eikonal equation, developed during the prior summer, into the lab's shared FSM++ GitHub repository with GoogleTest-based tests.
- Los Alamos National Laboratory — XCP Summer Workshop Student** Los Alamos, NM | Jun 2022 – Aug 2022
- Co-developed a second-order numerical fast sweeping method for the Eikonal equation.
 - Achieved 70% better accuracy than the first-order solution for a single-point initialization in a constant field.
 - Implemented distributed-memory parallel fast sweeping methods for both first- and second-order solutions.
 - Demonstrated improved computational performance over the serial method, with weak scaling approaching ideal behavior.
- Florida State University — Teaching Assistant** Tallahassee, FL | Aug 2020 – May 2026
- Taught Introduction to Astronomy laboratory sections of 20–30 students.
 - Delivered 15–20 minute introductory lectures to guide students through each laboratory.
 - Guided students through astronomical concepts spanning spectroscopy, telescopes, and observational methods.
 - Connected laboratory concepts to their use in contemporary astronomical research.
- College Advising Corps — Matriculation Coach** Providence, RI | Summer 2019
- Successfully prevented summer melt for 42 recent Rhode Island high school graduates from underserved communities.
 - Developed an executable program that generated recommended majors for students at each Rhode Island college.
- Brown University Academic Support Services — Academic Coach** Providence, RI | Aug 2019 – May 2020
- Coached college students one-on-one in time management, study strategies, and scheduling.
 - Collaborated with the Dean of Academic Support Services to implement new workshops.
 - Led academic-skills workshops for 30+ participants, including presentation and exam preparation.

PUBLICATIONS † *First author or significant contributor*

1. † “*Little Red Dots as Shock-Powered High-Energy Neutrino Sources*” Mera, T.; Ashall, C.; Horiuchi, S.; et al. 2026, ApJ (submitted).
2. “*Can Little Red Dots Contribute To The Early Universe Cosmic Dust Budget?*” Ashall, C.; Naidu, R. P.; Torralba, A.; et al. 2026, ApJ (submitted).
3. † “*JWST Observations of SN 2023ixf III: Directly Tracing the Formation of Molecules and Dust from 285 to 385 Days After the Explosion*” Mera, T.; Medler, K.; Ashall, C.; et al. 2026, ApJ (in preparation).
4. “*Panchromatic JWST Observations and Models of the Dim Type Ia Supernova 2024vjm at 200 days.*” Baron, E.; DerKacy, J.; ... ; Mera, T.; et al; et al. 2026, ApJL.
5. † “*Unveiling Molecule Formation in Supernova 2022acko with JWST: Late Phase Infrared Spectroscopy*” Medler, K.; Mera, T.; Ashall, C.; et al. 2026, ApJ
6. “*The Production of Electron-Capture Elements in Thermonuclear Supernovae: Theory vs. Observations*” Shiber, S.; Hoefflich, P.; Mera, T.; et al. 2026, AIP.

7. “*Filling the Pockets: The Spherical Nature of 3D Deflagration in Thermonuclear Supernovae*” Shiber, S.; Hoefflich, P.; Mera, T.; et al. 2026, ApJL.
8. † “*Probing the 3D Structures of Supernovae through IR Signatures of CO and SiO*” Mera, T.; Hoefflich, P.; Ashall, C.; et al. 2026, ApJ.
9. † “*JWST Observations of SN 2024ggi II: NIRSpec Spectroscopy and CO Modeling at +285–385 Days Past the Explosion*” Mera, T.; Ashall, C.; et al. 2026, ApJ.
10. “*JWST Observations of SN 2023ixf II: The Panchromatic Evolution Between 250 and 720 Days After the Explosion*” Medler, K.; Ashall, C.; ... ; Mera, T.; et al. 2025, ApJ.
11. “*JWST Observations of SN 2024ggi I: Interpretation and Model Comparison of the Type II Supernova 2024ggi at 55 Days Past Explosion*” Baron, E.; Ashall, C.; ... ; Mera, T.; et al. 2025, ApJ.
12. “*JWST Observations of SN 2023ixf I: Completing the Early Multi-Wavelength Picture with Plateau-phase Spectroscopy*” DerKacy, J.; Ashall, C.; ... ; Mera, T.; et al. 2025, ApJ.
13. “*Near-Infrared Spectroscopy and Detection of Carbon Monoxide in the Type II Supernova SN 2023ixf*” Park, S. H.; Rho, S.; ... ; Mera, T.; et al. 2025, A&A.
14. “*A JWST Medium Resolution MIRI Spectrum and Models of the Type Ia Supernova 2021aefx at +415d*” Ashall, C.; Hoefflich, P.; ... ; Mera, T.; et al. 2024, ApJ.
15. “*JWST NIRSpec+MIRI Observations of the Nearby Type IIP Supernova 2022acko*” Shahbandeh, M.; Ashall, C.; ... ; Mera, T.; et al. 2024, ApJ.
16. “*Type Ia Supernovae in the Age of JWST: Finding the ‘Right’ Questions and the Path to Answer*” Hoefflich, P.; Fereidouni, E.; Mera, T. 2023, Elsevier.
17. “*Type Ia Supernovae Progenitor Properties and Their Host Galaxies*” Chakraborty, S.; Sadler, B.; ... ; Mera, T. B.; et al. 2023, AAS (submitted).
18. “*JWST MIRI/MRS Observations and Spectral Models of the Under-luminous Type Ia Supernova 2022xkq*” Derkacy, J.; Ashall, C.; ... ; Mera Evans, T. B.; et al. 2023, ApJ.
19. “*JWST Low-Resolution MIRI Spectral Observations of SN 2021aefx: High-density Burning in a Type Ia Supernova*” Derkacy, J.; Ashall, C.; ... ; Mera Evans, T. B.; et al. 2023, ApJL.
20. “*A JWST Near- and Mid-Infrared Nebular Spectrum of the Type Ia Supernova 2021aefx*” Kwok, L.; Jha, S.; ... ; Mera Evans, T. B.; et al. 2023, ApJL.
21. † “*A Second-order Distributed Memory Parallel Fast Sweeping Method for the Eikonal Equation*” Tro, S.; Mera Evans, T. B.; et al. 2022, JCP.
22. † “*Galactic Positrons from Thermonuclear Supernovae*” Mera Evans, T. B.; Hoefflich, P.; Diehl, R. 2021, ApJ.

PRESENTATIONS

- “*CO and SiO Dust from Core-Collapse Supernovae*” Astro Seminar, Florida State University, Tallahassee, Florida — Apr 19, 2023
- “*Detonation Waves & Probing the Progenitor Structure of Thermonuclear Supernovae by the Fe II Line*” Astro Seminar, Florida State University, Tallahassee, Florida — Sep 14, 2022
- “*Thermonuclear Supernovae as a Source of Galactic Positrons*” 88th Southeastern Section of the American Physical Society, Florida State University, Tallahassee, Florida — Nov 19, 2021
- “*Estrategias de Estudio y Administracion del Tiempo*” Education Department Conference Talk, Universidad Laica Eloy Alfaro de Manabi, Manta, Ecuador — May 26, 2021

HONORS

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| Anna Runyan Endowment Award, Florida State University Department of Physics | May 2022 |
| Fellow, American Physical Society Bridge Program | Aug 2020 – Aug 2022 |

SKILLS

- Computing:** Fortran, Python, C++, OpenMPI, Bash, LaTeX, Stellarium
- Languages:** Spanish (basic/intermediate)
- Interests:** Calisthenics, mountain biking, running